



CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®
ENVIRONMENTAL SCIENCE
AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

UNIT 2 – Paper 02

2 hours 30 minutes

13 MAY 2014 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions, TWO from each module.
2. Answer **ALL** questions.
3. Write your answers in the answer booklet provided.
4. The use of a silent, non-programmable, scientific calculator is permitted.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Copyright © 2012 Caribbean Examinations Council
All rights reserved.



MODULE 1

Answer BOTH questions.

1. (a) Figure 1 shows the percentage of farmers in a country that use environmentally sustainable practices.

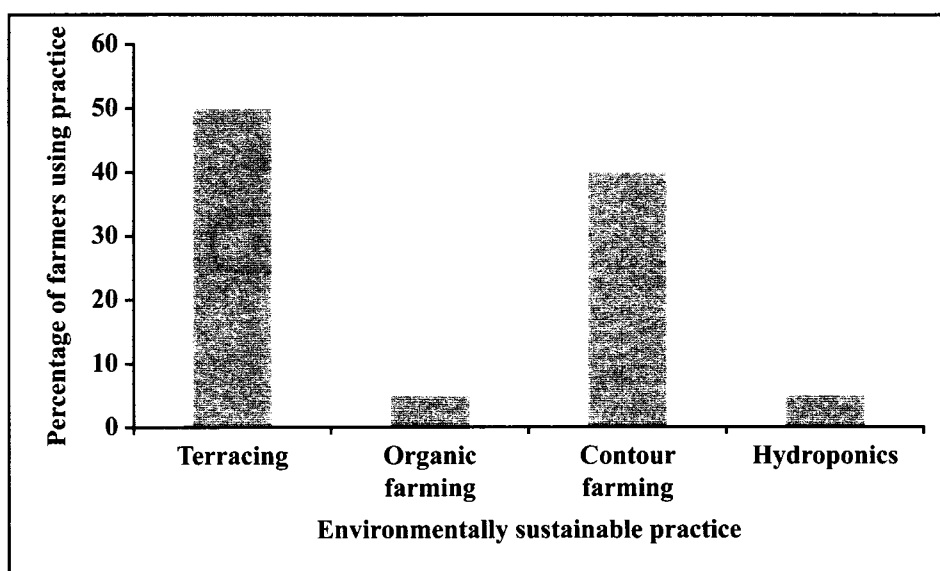


Figure 1. Percentage of farmers that use environmentally sustainable practices

- (i) State THREE deductions that could be made from the data presented in Figure 1. [3 marks]
 - (ii) Calculate the combined percentage of farmers who use the TWO LEAST popular practices. [2 marks]
 - (iii) Suggest TWO reasons why the practices identified in (a) (ii) above are not popular. [4 marks]
 - (iv) State TWO benefits that farmers derive from practising organic farming. [2 marks]
- (b) Explain, giving TWO reasons, why global warming should be of major concern to agriculture-based economies in the Caribbean. [6 marks]
- (c) Suggest THREE reasons why more farmers in the Caribbean are increasing their use of technology in agriculture. [3 marks]

Total 20 marks

GO ON TO THE NEXT PAGE

2. (a) Table 1 summarizes some characteristics of the agricultural systems on two farms, A and B.

TABLE 1: CHARACTERISTICS OF AGRICULTURAL SYSTEMS

Characteristics of Agricultural Systems	Farm A	Farm B
Use of agrochemicals	High	None
Yield per input	High	Low
Labour input	Low	High
Mechanization	High	None
Scale of operation	Large	Small

- (i) Which farm engages in subsistence agriculture? Using the information in Table 1, give FOUR reasons to justify your answer. **[5 marks]**
- (ii) Which farm is MOST likely to have a greater impact on the environment? Explain THREE reasons for your answer. **[7 marks]**
- (b) Explain why the practice of sustainable agriculture maintains ecological integrity. **[4 marks]**
- (c) Figure 2 depicts the percentage of pests that were eliminated using different pest control methods on Farm A.

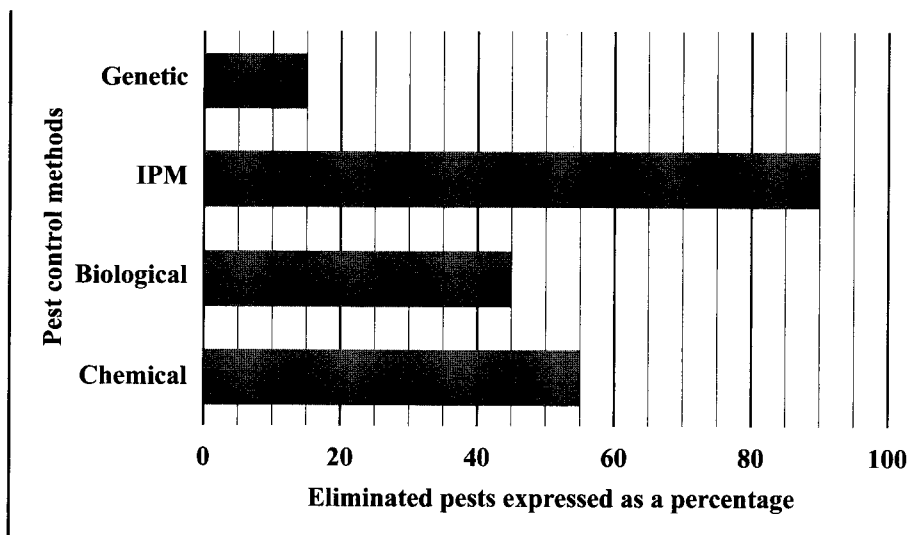


Figure 2. Percentage of pests eliminated using various pest control methods

- (i) What percentage of pests was eliminated using the LEAST effective method? **[1 mark]**
- (ii) Which method was the MOST effective in controlling pests on Farm A? **[1 mark]**
- (iii) Suggest TWO reasons why the method identified in (c) (ii) above may have been the MOST effective. **[2 marks]**

Total 20 marks

GO ON TO THE NEXT PAGE

MODULE 2

Answer BOTH questions.

3. (a) Figure 3 is an incomplete flow diagram illustrating the conversion of energy from the solar energy in sunlight to the kinetic energy in a moving, gasoline-powered vehicle. Copy this diagram into your answer booklet, and fill in the blank spaces on the diagram.

[3 marks]

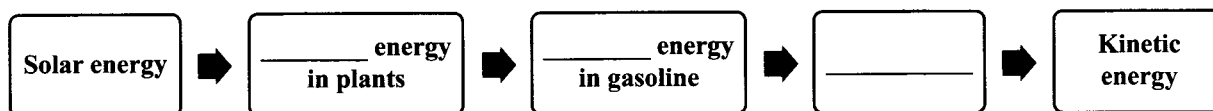


Figure 3. Incomplete flow diagram illustrating the conversion of energy from sunlight into kinetic energy in a moving vehicle

- (b) Table 2 illustrates specifications for two types of water heaters.

TABLE 2: SPECIFICATIONS FOR TWO TYPES OF WATER HEATERS

Specifications	Natural Gas	Electric
Size, l	190	190
Purchase cost, \$	2,230.00	1,850.00
Operational cost, \$/year	1,045.00	1,200.00
Energy factor (higher = more efficient)	0.90	0.58
Availability in stores	Rare	Common

- (i) Which type of water heater is cheaper to operate? [1 mark]
- (ii) What annual savings could be gained from owning the kind of water heater chosen in (b) (i) above? [1 mark]
- (iii) Consider the information provided in Table 2 and compare each water heater's performance based on economic and energy conservation issues. [6 marks]

GO ON TO THE NEXT PAGE

- (c) Figure 4 illustrates the trends in petroleum prices from 1980 to 2011. Carefully examine the data in the graph and answer the questions that follow.

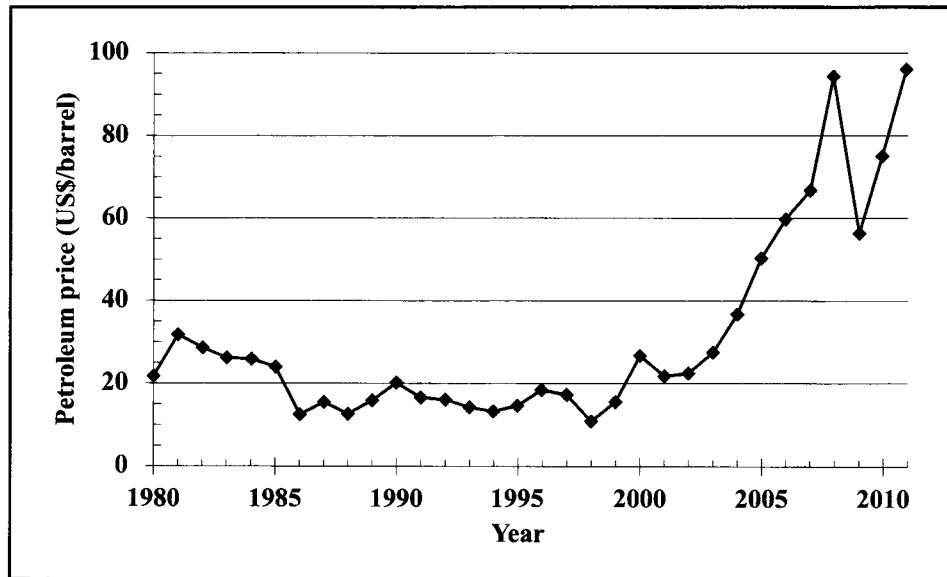


Figure 4. Petroleum prices from 1980 to 2011

- (i) Describe the trend illustrated in Figure 4. **[3 marks]**
- (ii) Give TWO reasons to explain how EACH of the following may produce the trend in petroleum prices seen in Figure 4:
- Technological limitations
 - Geographical restrictions
 - Reliability of supply
- [6 marks]**

Total 20 marks

4. Figure 5 illustrates the process of electricity generation by the combustion of natural gas. Study the figure and answer the questions that follow.

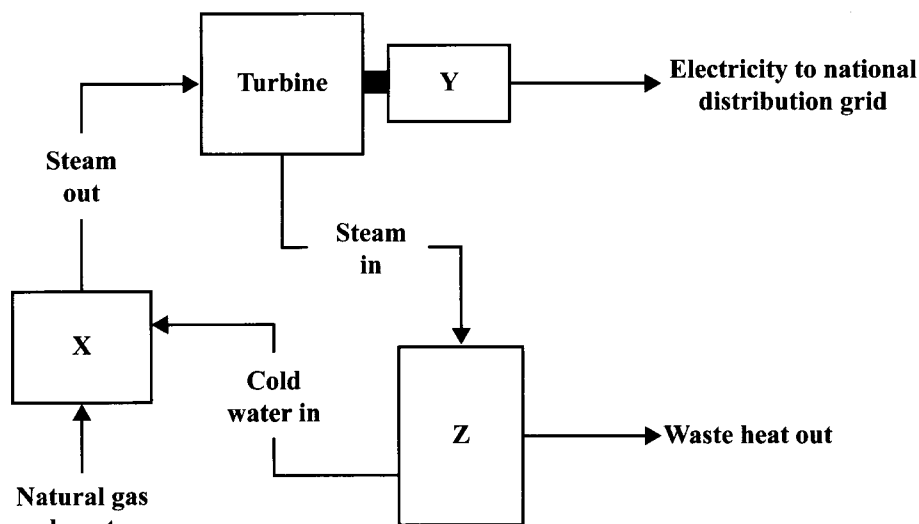


Figure 5. Flow diagram of the processes and units involved in the conventional generation of electricity

- (a) (i) Identify the units labelled X, Y and Z in Figure 5. [3 marks]
- (ii) Discuss in detail ONE environmental impact that is likely to result from the conventional generation of electricity from fossil fuels. [6 marks]
- (b) Table 3 presents the per capita electricity consumption and population sizes of a number of Caribbean countries in 2006. Study the table and answer the questions that follow.

TABLE 3: PER CAPITA ELECTRICITY CONSUMPTION AND POPULATION SIZE OF SELECTED CARIBBEAN COUNTRIES IN 2006

Country	Per Capita Electricity Consumption (Million BTU)	Population
Antigua and Barbuda	135	76 000
Jamaica	59.3	2 700 000
Saint Lucia	32.2	160 000
Montserrat	106.6	9 500

- (i) Rank the countries in order of DECREASING per capita electricity consumption. [1 mark]
- (ii) Use the data in Table 3 to determine which country has the SMALLEST total electricity consumption. [4 marks]
- (iii) Explain how EACH of the following can affect electricity generating demand:
- Economic cost
 - Government policies
- [6 marks]

Total 20 marks

GO ON TO THE NEXT PAGE

MODULE 3

Answer **BOTH** questions.

5. (a) A list of pollutants and their sources can be found below. Correctly match the pollutants with their sources. **[3 marks]**

Pollutants: Ammonia; Leachate; Carbon dioxide

Sources: Forest fires; Industrial activities; Landfills

- (b) Two large ponds in a Caribbean country are being affected by effluents contaminated with high levels of nitrate. Table 4 provides information about the characteristics of the two ponds and the effluents that enter them. Use the data in Table 4 to answer the questions that follow.

**TABLE 4: CHARACTERISTICS OF TWO LARGE, NITRATE-AFFECTED PONDS
IN A CARIBBEAN COUNTRY**

Characteristic	Pond A	Pond B
Volume, m ³	1 500 000	5 650 000
Effluent flow rate into the pond, m ³ /day	4.5	2.0
Nitrate concentration in effluent, mg/l	350	350
Nitrate removal rate from the pond, mg/l/day	20	250
Average nitrate concentration in the pond, mg/l	2 550	400

- (i) Explain why the average nitrate concentrations in the two ponds are so different, even though the concentration of nitrate entering each pond is the same. **[3 marks]**
- (ii) State the name of the **MOST** likely environmental impact that will occur in Ponds A and B **AND** discuss how this impact will affect the ponds. **[6 marks]**
- (iii) Suggest **ONE** measure that could be taken to mitigate the environmental impact stated in (b) (ii) above. **[1 mark]**
- (iv) Would the measure suggested in (b) (iii) above be effective? Give **ONE** reason for your answer. **[2 marks]**

GO ON TO THE NEXT PAGE

- (c) Not all pollutants are generated by humans; sometimes natural processes can result in water pollution. Figure 6 illustrates the result of an experiment to determine how rapidly the turbidity of a river changed after a short rainfall event. Study the figure and answer the questions that follow.

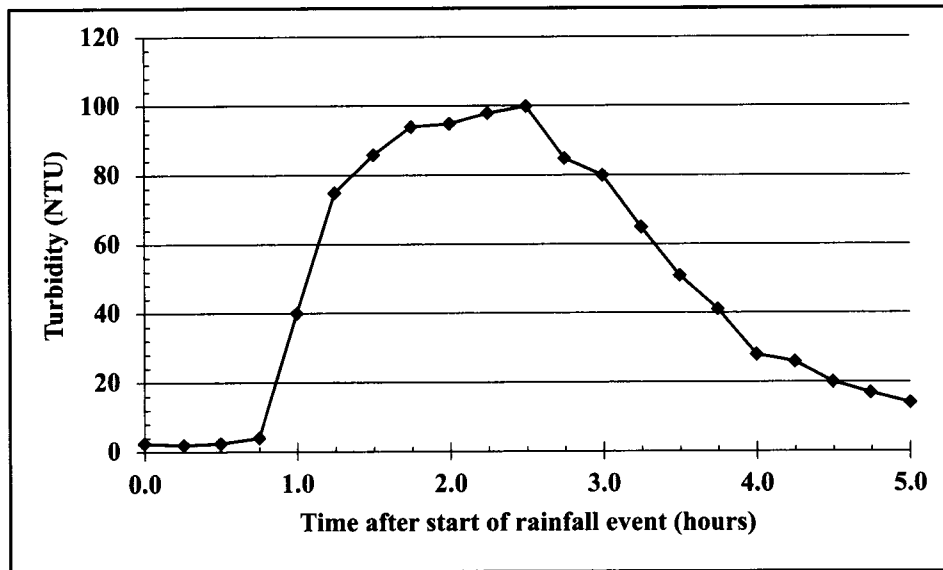


Figure 6. Results of turbidity monitoring experiment

- (i) How long did it take for the river to show the effects of the rainfall event? **[1 mark]**
- (ii) When was turbidity at its maximum? **[1 mark]**
- (iii) The following statement was recorded in the experimenter's laboratory notebook.
"Five hours is a suitable experimental time for the river to achieve its pre-rainfall event turbidity."
Assess this statement. **[3 marks]**

Total 20 marks

GO ON TO THE NEXT PAGE

6. The flowchart in Figure 7 illustrates a pathway for the movement of a pesticide in the environment. Some of the arrows are in the WRONG direction. Study the diagram and answer the questions that follow.

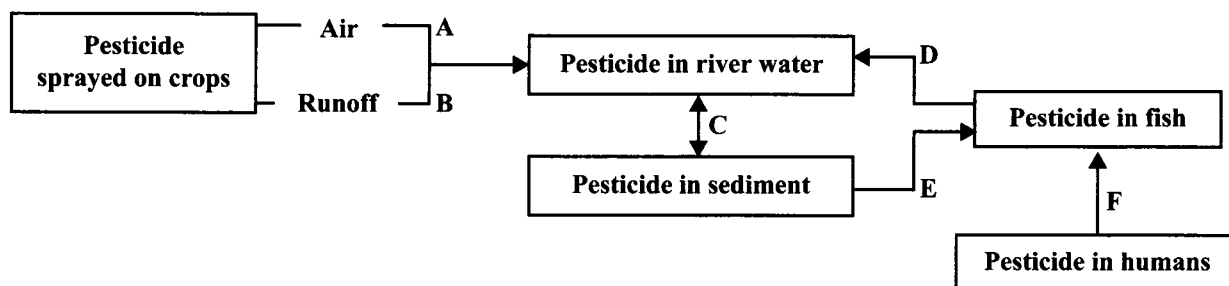


Figure 7. Movement of a pesticide through the environment.

- (a) Identify which of the arrows labelled A – F in Figure 7 are INCORRECTLY placed. [2 marks]
- (b) The combustion of fossil fuels is known to cause atmospheric pollution. However, this is just the last stage of pollutant generation by fossil fuel use. Pollution can also occur from the extraction stage, the transportation stage or the processing stage for fossil fuels. Choose ONE of these stages and explain why it may also lead to environmental pollution. [3 marks]
- (c) (i) Using ONLY a suitably annotated diagram that shows the movement of energy, explain the ‘greenhouse effect’. [5 marks]
- (ii) What is the Kyoto Protocol? [2 marks]
- (iii) Currently, the Kyoto Protocol has not been very successful. Suggest ONE reason why this is so. [3 marks]

GO ON TO THE NEXT PAGE

- (d) Figure 8 illustrates the levels of CFC-11 in the atmosphere over the period 1977-2011. Study the figure and answer the questions that follow.

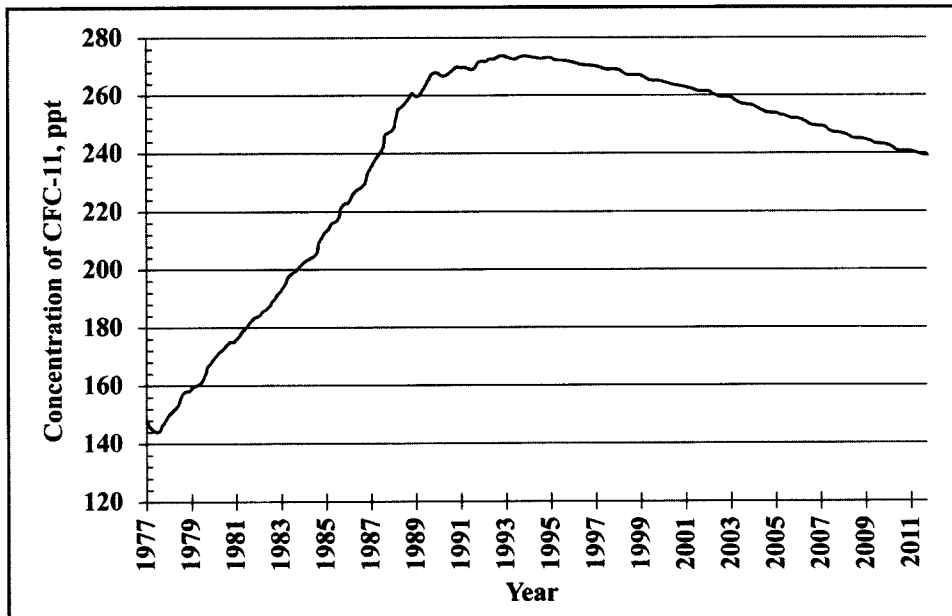


Figure 8. Atmospheric CFC-II levels from 1977 to 2011

- (i) What was the annual rate of increase of atmospheric CFC-11 from 1979 to 1989? **[3 marks]**
- (ii) The Montreal Protocol first entered into force sometime in the 1980s. During what period did the Montreal Protocol begin to have an effect on the atmospheric levels of CFC-11? **[2 marks]**

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.